

Technology Stack

Date	1 July 2026
Team ID	xxxxxx
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML, CSS, JavaScript	Responsive web interface for entering weather parameters and displaying flood prediction results.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python, Flask	Hosts the machine learning model, processes user requests, and returns flood prediction results.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	Flood Dataset (CSV/Excel), Joblib Saved Model	Stores the training dataset and serialized trained model used for prediction.
4	Cloud / Hosting / Deployment	IBM Cloud	Provides scalable cloud deployment and public accessibility for the application.

	<i>(e.g., AWS, Azure, Heroku, Docker)</i>		
5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	GitHub	Supports version control, collaboration, and project source code management.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Scikit-learn, XGBoost, NumPy, Pandas, Matplotlib	Used for machine learning, data preprocessing, visualization, model training, and evaluation